JIE YUAN

FRAeS · CEng · FHEA

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RESEARCH PROFILE

I develop robust computational methods for the design and optimisation of complex, large-scale nonlinear aerospace systems, spanning nonlinear structural dynamics and friction-damped turbomachinery, dynamic control via novel functional materials and interfaces, and non-deterministic (Bayesian/AI-assisted) approaches to robust optimisation and identification. I hold a Royal Academy of Engineering / Leverhulme Trust Research Fellowship, have secured over £2.1M in research funding as PI and Co-I, and have authored 39+ peer-reviewed publications (1,163 citations, h-index 21, i10-index 31 — Google Scholar, as of Jul 2026). I currently supervise six PhD researchers and welcome enquiries from prospective doctoral students.

01 EMPLOYMENT HISTORY

- **Jul 2023 – present —** Assistant Professor in Aerospace Design and Optimization, Department of Aeronautics and Astronautics, University of Southampton, UK
- **Dec 2020 – Jun 2023 —** Lecturer in Aerospace Engineering, Department of Mechanical and Aerospace Engineering, University of Strathclyde, Glasgow, UK
- **Apr 2017 – Nov 2020 —** Research Associate, Vibration University Technology Centre, Department of Mechanical Engineering, Imperial College London, UK — Projects: EPSRC SYSDYMAT (2019–2022) and ATI GEMinIDS (2017–2019)
- **Oct 2015 – Mar 2017 —** Research Engineer, Airbus UK & Cranfield University, Bristol, UK

02 EDUCATION

- **Jul 2012 – Feb 2016 —** PhD in Aerospace Engineering, University of Bristol, UK — supervised by Prof. Fabrizio Scarpa
- **Oct 2010 – Feb 2012 —** MSc in Integrated Aerospace System Design (ranked top 3 of cohort), University of Bristol, UK
- **Sep 2006 – Jul 2010 —** BEng in Mechanical Engineering (First Class, ranked 5/200), Ningbo University, Zhejiang, China

03 HONOURS & AWARDS

- Fellow of the Royal Aeronautical Society (2025)
- Research Fellowship, Royal Academy of Engineering / Leverhulme Trust (2023)
- Fellow of the Higher Education Academy (2022)
- Chartered Engineer, UK Engineering Council (2021)
- Best Paper Award, Technical Division on Nonlinear Structures and Systems, IMAC (2021)

04 PROFESSIONAL MEMBERSHIPS & CERTIFICATIONS

- Committee Member, Propulsion Specialist Group, Royal Aeronautical Society
- Scientific Committee Member, International Structural Joint Community
- Youth Editorial Board Member, Theoretical and Applied Mechanics Letters (Elsevier)
- Member, American Society of Mechanical Engineers (ASME)
- Member, Society for Experimental Mechanics (SEM)

05 RESEARCH FUNDING

Fellowships & Research Grants (PI unless noted)

- **Co-PI (1 of 2) —** DDT: Digital Design Transformation — Innovate UK

  2025–2028 | Value: £1.62M (shared)

- **PI —** Invariant-manifold-based reduced-order modelling for nonlinear aeroelastic systems — EPSRC Postdoctoral Fellowship (Dr Michael McGurk)

  2025–2027 | Value: n/d

- **PI —** Multi-scale Friction Interface Design for Resilient Dynamical Systems — Royal Academy of Engineering Fellowship

  2023–2024 | Value: £58K

- **PI —** Controlling Limit Cycle Oscillations for Complex Aerospace Systems — Royal Society Research Grant

  2024–2025 | Value: £20K

- **PI —** Multi-functional Nonlinear Tuned Vibration Energy Harvester for Intelligent Rotating Systems — Royal Society International Exchanges

  2022–2025 | Value: £12K

- **PI —** Uncertainty Quantification of Novel Aeroelastic Systems using Advanced Composite Materials — EPSRC Impact Acceleration Award

  2022 | Value: £15K

- **PI —** Multi-scale Thermomechanical Analysis of Friction-Involved Complex Dynamic Systems — Royal Society of Edinburgh — Small Research Award

  2021–2022 | Value: £5K

- **PI —** Uncertainty Quantification for Nonlinear Dynamic Systems with Friction Interface — Royal Society of Edinburgh — Saltire Workshop Award

  2022–2023 | Value: £10K

PhD Studentships (PI or Awardee)

- **PI —** AI-enabled Stochastic Load Identification due to Fan Blade-Off — EPSRC iCASE with Rolls-Royce plc, University of Southampton

  2025–2029 | Value: £160K

- **PI —** Robust Nonlinear Aeroelasticity Tailoring using Composite Materials — EPSRC DTP Research Excellence Award, University of Strathclyde

  2021–2025 | Value: £85K

- **PI —** Mutual Effects between Friction and Vibration across Scale and Physics — Strathclyde International Partnership, University of Strathclyde

  2022–2025 | Value: £60K

- **Awardee —** Reduced-Order Modelling of Mistuned Bladed Disc Systems — Rolls-Royce Studentship, University of Bristol

  2012–2016 | Value: £60K

Funding summary (PI + Co-I, combined): ~£120K research grants (PI) + ~£365K PhD studentships (PI/Awardee) + £1.62M consortium grant (Co-PI, 1 of 2) = **~£2.1M total**. Figures are drawn directly from the grants listed above; the £1.62M consortium award is shared with a co-investigator and one fellowship value was not disclosed in the source CV.

06 INVITED TALKS

- "Friction damping for turbomachinery: a comprehensive review of modelling, design strategies, and testing capabilities"

  University of Cambridge, UK — Mar 2025

- "Data-driven Bayesian identification for complex aeroelastic systems"

  Imperial College London, UK — Oct 2024

- "Influence of mesoscale friction interface geometry on nonlinear dynamics"

  INRIA, Rennes, France — 26 Oct 2022

- "High fidelity modelling of jointed structures with friction interface in aero engine applications"

Digital Twins Workshop on Numerical and Experimental Studies of Structural Joints, University of Swansea — Jun 2021

- "An overview of reduced order modelling techniques for dynamic analysis of jointed structures with localized nonlinearities due to contact friction"

  23rd Blade Mechanics Seminar, Winterthur, Switzerland — Sep 2018

07 ACADEMIC ENGAGEMENT & SERVICE

- EPSRC Peer Review College (2024–present)
- Session Chair, Computational Engineering Design, Annual Rolls-Royce Doctoral Symposium (2024–2026)
- Scientific Committee Member, conference on "Recent Advances in Non-Smooth Dynamics", Exeter, UK (Dec 2025)
- Workshop Organiser, "Multiscale Structural Dynamics of Uncertain Frictional Interfaces" (Jun 2023)
- Lead Guest Editor, ASCE-ASME Journal of Risk and Uncertainty in Engineering Systems, Part B — Special Issue on "Uncertainty Quantification & Management in Nonlinear Dynamical Systems in Aerospace Engineering" (2022–2023)
- Lead Guest Editor, Applied Mechanics (ISSN 2673-3161) — Special Issue on "Vibrations and Friction: Theoretical, Experimental and Numerical Studies" (2021–2024)
- Session Chair, 3rd & 4th International Nonlinear Dynamics Conference (Jun 2023 & 2025)
- Peer Reviewer (8–10 papers/year): Mechanical Systems and Signal Processing, AIAA Journal, Journal of Sound and Vibration, Nonlinear Dynamics, Proceedings of the Royal Society A
- Departmental Seminar & Conference Organiser, Department of Aeronautics and Astronautics, University of Southampton

08 TEACHING ACTIVITIES

- **Digital Aerospace —**

  UG · Lecturer & Co-module Lead · University of Southampton · 2024–present · ~250 students
- **Statics and Dynamics —**

  PG · Lab Lead · University of Southampton · 2024–present · ~250 students
- **Aeroelasticity —**

  PG · Lecturer · University of Southampton · 2025–present · ~10 students
- **Aerospace Propulsion —**

  UG · Lecturer & Module Lead · University of Strathclyde · 2020–2023 · ~120 students
- **Aero Design 1 —**

  UG · Lecturer · University of Strathclyde · 2020–2023 · ~150 students
- **MSc Advanced Mechanical Engineering —**

  Programme Director · University of Strathclyde · 2020–2023 · ~40 students
- **Postgraduate Internal & External Examiner —**

  University of Strathclyde and University of Bristol

09 PHD SUPERVISION

Current

- **Mingzhi Huang —** Main Supervisor · University of Southampton · 2025–present

  Project: Design and optimisation of nature-inspired architected structures for passive dynamic control
- **Giovanna Vilela —** Main Supervisor · University of Southampton · 2025–present

  Project: AI-assisted robust aero-engine structural analysis due to fan blade-off
- **Peiyu Wang —** Main Supervisor · University of Southampton · 2024–present

  Project: Robust data-driven modelling for complex nonlinear and high-dimensional dynamic systems
- **Thomas Tsing —** Co-supervisor · University of Southampton · 2024–present

Project: Convolutional generative adversarial networks for rapid engineering design

- **Ahmad Algara —** Main Supervisor · University of Strathclyde · 2022–present

  Project: Multi-scale modelling for friction-induced vibration systems

- **Harry Leitch —** Main Supervisor · University of Strathclyde · 2022–present

  Project: Robust optimisation of tow-steered composites for aerospace applications

Completed

- **Dr Michael McGurk —** Main Supervisor · University of Strathclyde · 2021–2025 (completed)

  Project: A robust data-driven Bayesian approach for complex nonlinear aeroelastic system identification

- **Dr Yekai Sun —** Co-supervisor · Imperial College London · 2017–2021 (completed)

  Project: Nonlinear modal analysis for blisks with friction dampers

10 SELECTED PUBLICATIONS

Google Scholar: 1,163 citations · h-index 21 · i10-index 31 (as of Jul 2026).

1. Algara, A., Cabboi, A., & **Yuan, J.** (2026). Passive control through surface roughness of local stability boundaries for friction-induced vibration. International Journal of Non-Linear Mechanics, 105419. https://doi.org/10.1016/j.ijnonlinmec.2026.105419
2. Wang, P., Toal, D., & **Yuan, J.** (2026). CNN-assisted robust design of an intentionally mistuned frictional damper for bladed-disk systems. ASME Journal of Turbomachinery (accepted).
3. Yang, D., Lu, Z., **Yuan, J.** et al. (2026). An event-driven time-domain sensitivity method for optimizing parameters of vibro-impact NES cells to suppress multi-mode vibration. Journal of Sound and Vibration. https://doi.org/10.1016/j.jsv.2026.119664
4. Shao, K., Wang, T., **Yuan, J.**, Dong, X., & Yang, J. (2026). Vibration power flow and dynamic interactions in nonlinear mistuned bladed disk system. International Journal of Mechanical Sciences, 111190. https://doi.org/10.1016/j.ijmecsci.2026.111190
5. Huang, M., **Yuan, J.**, Leitch, H., & Scarpa, F. (2025). Analysis and optimisation of cellular Kirigami wingbox (CKW) structures for enhanced aeroelastic performance. The Aeronautical Journal, 1–17. https://doi.org/10.1017/aer.2025.10073
6. Leitch, H. J., Stodieck, O., & **Yuan, J.** (2025). The influence of coupled thickness variation in the aeroelastic response of continuous tow sheared composite wing. Composite Structures, 119706. https://doi.org/10.1016/j.compstruct.2025.119706
7. McGurk, M., & **Yuan, J.** (2025). Prediction and validation of aeroelastic limit cycle oscillations using harmonic balance methods and Koopman operator. Nonlinear Dynamics, 113(22), 30841–30868. https://doi.org/10.1007/s11071-025-11065-8
8. Zhong, Y., Ma, X., **Yuan, J.**, & Zhang, Z. (2024). Electromechanical modelling and vibration control analysis of cyclic symmetric structures with piezoelectric vibration absorber. Journal of Vibration & Control. https://doi.org/10.1177/10775463241299858
9. Wang, X., **Yuan, J.**, Szydlowski, M., & Schwingshackl, C. (2024). Full-field measurement of lightweight nonlinear structures using 3D SLDV. AIAA Journal. https://doi.org/10.2514/1.J064240
10. **Yuan, J.**, Gastaldi, C., Denimal Goy, E., & Chouvion, B. (2024). Friction damping for turbomachinery: a comprehensive review of modelling, design strategies, and testing capabilities. Progress in Aerospace Sciences, 147, 101018. https://doi.org/10.1016/j.paerosci.2024.101018
11. McGurk, M., Lye, A., Renson, L., & **Yuan, J.** (2024). Data-driven Bayesian inference for stochastic model identification of nonlinear aeroelastic systems. AIAA Journal, 62(5), 1889–1905. https://doi.org/10.2514/1.J063611
12. **Yuan, J.**, Szydlowski, M., & Wang, X. (2024). An optimal sparse sensing approach for scanning point selection and response reconstruction in full field structural vibration testing. Mechanical Systems and Signal Processing, 212, 111298. https://doi.org/10.1016/j.ymssp.2024.111298
13. Tufekci, M., Sun, Y., **Yuan, J.**, Maharaj, C., Liu, H., Dear, J. P., & Salles, L. (2024). Analytical vibration modelling and solution of bars with frictional clamps. Journal of Sound and Vibration. https://doi.org/10.1016/j.jsv.2024.118307
14. McGurk, M., **Yuan, J.**, & Stodieck, O. (2023). Probabilistic aeroelastic analysis of high-fidelity composite aircraft wings with manufacturing variability. Composite Structures. https://doi.org/10.1016/j.compstruct.2023.117794
15. Li, Q., Ainsworth, O., Allegri, G., **Yuan, J.**, & Scarpa, F. (2023). Assessing the mechanical and static aeroelastic performance of cellular Kirigami wingbox designs. Aerospace Science and Technology, 143, 108716. https://doi.org/10.1016/j.ast.2023.108716
16. Zhang, W., Neville, R., Zhang, D., **Yuan, J.**, Scarpa, F., & Lakes, R. (2023). Bending of kerf chiral fractal lattice metamaterials. Composite Structures, 318, 117068. https://doi.org/10.1016/j.compstruct.2023.117068

17. Peddakotla, S. A., **Yuan, J.**, Minisci, E., Vasile, M., & Fossati, M. (2023). A numerical approach to evaluate temperature-dependent peridynamics damage model for destructive atmospheric entry of spacecraft. The Aeronautical Journal, 127(1309), 398–427. https://doi.org/10.1017/aer.2022.69

18. **Yuan, J.**, Salles, L., Nowell, D., & Schwingshackl, C. (2023). Influence of mesoscale friction interface geometry on the nonlinear dynamic response of large assembled structures. Mechanical Systems and Signal Processing, 187, 109952. https://doi.org/10.1016/j.ymssp.2022.109952

19. Ma, X., Zhong, Y., Cao, P., **Yuan, J.**, & Zhang, Z. (2023). Uncertainty quantification and sensitivity analysis for the self-excited vibration of a spline-shafting system. Journal of Risk and Uncertainty in Engineering Systems, Part B: Mechanical Engineering. https://doi.org/10.1115/1.4063069

20. Yekai, S., Yaguang, W., Xing, W., Yu, F., **Yuan, J.**, & Dayi, Z. (2022). Review of the damped nonlinear normal mode in the analysis of friction dampers of blades/blisks. Journal of Aerospace Power, 37(10), 2167–2187. https://doi.org/10.13224/j.cnki.jasp.20220264

21. Wang, X., Szydlowski, M., **Yuan, J.**, & Schwingshackl, C. (2021). A multi-step interpolated-FFT procedure for full-field nonlinear modal testing of turbomachinery components. Mechanical Systems and Signal Processing, 169. https://doi.org/10.1016/j.ymssp.2021.108771

22. Zhu, Y.-P., **Yuan, J.**, Lang, Z. Q., Schwingshackl, C. W., Salles, L., & Kadirkamanathan, V. (2021). The data driven surrogate model based dynamic design of aero-engine fan systems. Journal of Engineering for Gas Turbines and Power, 143(10), 101006. https://doi.org/10.1115/1.4049504

23. **Yuan, J.**, Sun, Y., Schwingshackl, C., & Salles, L. (2021). Computation of damped nonlinear normal modes for large scale nonlinear systems in a self-adaptive modal subspace. Mechanical Systems and Signal Processing, 162, 108082. https://doi.org/10.1016/j.ymssp.2021.108082

24. Sun, Y., **Yuan, J.**, Vizzaccaro, A., & Salles, L. (2021). Comparison of different methodologies for the computation of damped nonlinear normal modes and resonance prediction of systems with non-conservative nonlinearities. Nonlinear Dynamics, 104, 3077–3107. https://doi.org/10.1007/s11071-021-06567-0

25. Sun, Y., **Yuan, J.**, Denimal, E., & Salles, L. (2021). Nonlinear modal analysis of frictional ring damper for compressor blisk. Journal of Engineering for Gas Turbines and Power, 143(3). https://doi.org/10.1115/1.4049761

26. **Yuan, J.**, Fantetti, A., Denimal, E., Bhatnagar, S., Pesaresi, L., Schwingshackl, C., & Salles, L. (2021). Propagation of friction parameter uncertainties in the nonlinear dynamic response of turbine blades with underplatform dampers. Mechanical Systems and Signal Processing, 156, 107673. https://doi.org/10.1016/j.ymssp.2021.107673

27. **Yuan, J.**, Schwingshackl, C., Wong, C., & Salles, L. (2021). On an improved adaptive reduced-order model for the computation of steady-state vibrations in large-scale non-conservative systems with friction joints. Nonlinear Dynamics, 103, 3283–3300. https://doi.org/10.1007/s11071-020-05890-2

28. Sun, Y., Vizzaccaro, A., **Yuan, J.**, & Salles, L. (2021). An extended energy balance method for resonance prediction in forced response of systems with non-conservative nonlinearities using damped nonlinear normal mode. Nonlinear Dynamics, 103, 3315–3333. https://doi.org/10.1007/s11071-020-05793-2

29. Sun, Y., **Yuan, J.**, Pesaresi, L., Denimal, E., & Salles, L. (2020). Parametric study and uncertainty quantification of the nonlinear modal properties of frictional dampers. Journal of Vibration and Acoustics, 142(5). https://doi.org/10.1115/1.4046953

30. **Yuan, J.**, Salles, L., El Haddad, F., & Wong, C. (2020). An adaptive component mode synthesis method for dynamic analysis of jointed structure with contact friction interfaces. Computers and Structures, 229, 106177. https://doi.org/10.1016/j.compstruc.2019.106177

31. Xie, W., Yang, Y., Meng, S., Peng, T., **Yuan, J.**, Scarpa, F., Xu, C., & Jin, H. (2019). Probabilistic reliability analysis of carbon/carbon composite nozzle cones with uncertain parameters. Journal of Spacecraft and Rockets, 56(6), 1765–1774. https://doi.org/10.2514/1.A34392

32. **Yuan, J.**, El-Haddad, F., Salles, L., & Wong, C. (2019). Numerical assessment of reduced order modeling techniques for dynamic analysis of jointed structures with contact nonlinearities. Journal of Engineering for Gas Turbines and Power, 141(3), 031027. https://doi.org/10.1115/1.4041147

33. **Yuan, J.**, Scarpa, F., Titurus, B., Allegri, G., Patsias, S., & Rajasekaran, R. (2017). Novel frame model for mistuning analysis of bladed disk systems. Journal of Vibration and Acoustics, 139(3). https://doi.org/10.1115/1.4036110

34. **Yuan, J.**, Scarpa, F., Allegri, G., Titurus, B., Patsias, S., & Rajasekaran, R. (2017). Efficient computational techniques for mistuning analysis of bladed discs: a review. Mechanical Systems and Signal Processing, 87(Part A), 71–90. https://doi.org/10.1016/j.ymssp.2016.09.041

35. **Yuan, J.**, Allegri, G., Scarpa, F., Patsias, S., & Rajasekaran, R. (2016). A novel hybrid Neumann expansion method for stochastic analysis of mistuned bladed discs. Mechanical Systems and Signal Processing, 72–73, 241–253. https://doi.org/10.1016/j.ymssp.2015.11.011

36. Ma, X., Scarpa, F., Peng, H.-X., Allegri, G., **Yuan, J.**, & Ciobanu, R. (2015). Design of a hybrid carbon fibre/carbon nanotube composite for enhanced lightning strike resistance. Aerospace Science and Technology, 47, 367–377. https://doi.org/10.1016/j.ast.2015.10.002

37. **Yuan, J.**, Allegri, G., Scarpa, F., Rajasekaran, R., & Patsias, S. (2015). Novel parametric reduced order model for aeroengine blade dynamics. Mechanical Systems and Signal Processing, 62–63, 235–253. https://doi.org/10.1016/j.ymssp.2015.02.015

38. **Yuan, J.**, Allegri, G., Scarpa, F., Rajasekaran, R., & Patsias, S. (2015). Probabilistic dynamics of mistuned bladed disc systems using subset simulation. Journal of Sound and Vibration, 350, 185. https://doi.org/10.1016/j.jsv.2015.04.015

39. **Yuan, J.**, Allegri, G., & Scarpa, F. (2013). Buffeting mitigation using carbon nanotube composites: a feasibility study. Proceedings of the IMechE Part G: Journal of Aerospace Engineering, 227(9), 1425–1440. https://doi.org/10.1177/0954410012461986